ALM - 4

Case study :- Social media :-

→ Graph DB for relationships.

→ Document DB for posts.

→ key-value for sessions.

Data base :- Storing of data in a systematic manner. like row, columns etc.

Types of Databases :-

* Graph Data base
* Document Database
* key value Database

→ In Social media handling, it uses these 3 databases for storing its different types of data.

* Graph Database for relationships :-

→ Social media is all about Relationships :-

* Users follow other users.
* Users like posts
* Users Comment on post
* Users join groups.
* Mutual friends / Suggestions.

→ These are complex, interconnected relationships that are inefficient in traditional relational databases.

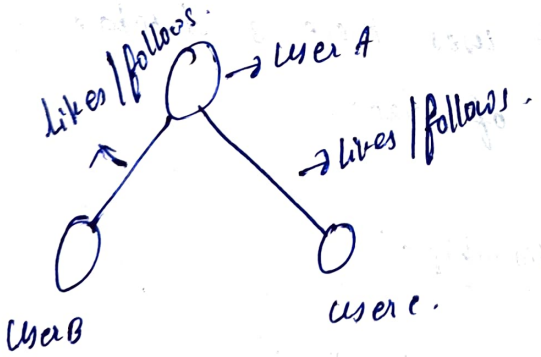
So, that we are using graph databases like:-

* Neo4j
* Amazon Neptune
* Arango DB.

Example of Graph Structure:-

→ Nodes → Users, Posts, groups.

→ Edges → follows, friends-with, likes, Member-of



Use Cases:-

→ friend recommendations.

→ People you may know.

→ Mutual friends.

→ shortest connection path

→ influencer detection etc.

Document database → Managing Posts & Content :-

→ Social media posts are semi-structured.

→ flexible.

→ frequently changing

→ May contain text, images, hastags, comments etc.

Ex :-

→ Mango DB

→ Couch DB

→ firebase firestore.

Use cases :-

→ User feed

→ Comments

→ Reactions

→ Stories

→ profile information.

Key-value Database - Managing Sessions & Caches :-

Session data requires :-

→ fast read/writes

→ Temp Storage

→ Expiration.

→ Simple Key lookup.

Ex :-

→ Redis

→ memcached

→ Dynamo DB.

Use cases :-

→ login Sessions.

→ Authentication tokens.

→ OTP Storage

→ Caching Trending posts. etc.

Real-time Working Scenario :-

→ User log in → Session Stored in Redis.

→ User opens feed

↳ Graph DB finds people they follow

→ Document DB fetches posts.

→ User likes post →

Graph DB stores likes relationship

Document DB increments like count.

→ Trending posts cached in Redis.

Benefits of this Design:-

→ High Scalability.

→ flexible Schema

→ fast relationship queries.

→ fast session handlings

→ Better performance at large scale.

System Architecture:-

Client App
(web / mobile App)

↓

load Balancer

↓

Auth Service → Key value DB (Sessions)

User Service → Graph DB (Relations)

Post Service → Document DB (Posts)

↓

feed Service (Combines Data).